

Potential Out-of-State Tag Holder Review — Method

A decision-support scorer for the 12,000 challenge cases. For each case it produces a review classification, three class probabilities, and a review-priority value, at both T0 and T1.

Everything below is reproducible from the code in this folder: `diagnostics.py` prints the evidence for each structural claim, `model_selection.py` and `audit_*.py` print the comparisons behind each modelling choice, `audit_honest.py` prints the unbiased estimate, and `predict.py` runs the whole thing.

1. What the data actually is

Nothing in the package names a join key between `candidate_records.csv` and the six evidence feeds. The feeds carry only names (and a date of birth on the licence feed). Three properties of the data drive every design decision that follows.

Each feed holds one record set per candidate. The address, licence, title and external feeds each hold 4.01 rows per candidate; the work-location feed and the T1 update stream each hold ~2.0. These are not coincidences across five tables — the evidence was generated per candidate, so linking is an *assignment* problem, not an independent per-row lookup.

Names are corrupted on a 3-good-plus-1-replaced pattern. The title feed is the one place this can be verified without assuming an answer, because `vehicle_ref` groups a vehicle's history independently of who is named on it. Clustering owner names inside each 4-row vehicle gives a 3 + 1 split for 98% of vehicles: three rows carrying one name, lightly corrupted (truncated to an initial, one substituted character, an adjacent transposition, a stray space), and one row carrying a name with no relationship to the other three. Grouping the licence feed by date of birth reproduces the same 3 + 1 pattern, so it is the generator's noise model rather than a quirk of one table.

The replaced name is a persistent alias, not fresh noise. That is what makes the fourth row recoverable in feeds beyond titles; see section 2.

The three classes are ordered. In the 300 development cases, no case moves from `review_not_warranted` to `review_warranted` between phases (0 cases), and one moves the other way. Every other move passes through `insufficient_evidence`. That is the signature of a single latent score with two thresholds, not three unrelated categories.

T0 \ T1	insufficient	not warranted	warranted
insufficient_evidence	71	22	12
review_not_warranted	20	85	0
review_warranted	12	1	77

Reading the axis against the evidence: cases whose *recent* records point to Delaware are labelled `review_warranted`; cases whose recent records point to another state are `review_not_warranted`; cases in between are `insufficient_evidence`. That matches the operational question — a vehicle whose owner now appears to be operating in Delaware is the one a staff member should look at.

2. Identity resolution

Resolution runs in three passes.

Pass 1 — titles, grouped by `vehicle_ref`. Rows are clustered by name inside each vehicle; each cluster is assigned to the candidate it agrees with best; the replaced-name singleton is attached to the vehicle's dominant owner at a reduced weight. This recovers the fourth row that name matching cannot reach: 38,582 to 47,581 linked rows, and the feed's Spearman rho against the ordered label rises from +0.25 to +0.37.

Pass 2 — the alias. The replaced name is not drawn fresh per row. Taking the odd-one-out name from each clean 3 + 1 vehicle and looking for it in the other feeds:

	replaced name found	recombined-name control
address feed	43.3%	3.8%
external feed	43.6%	3.9%
licence feed	31.9%	3.8%

An 11x enrichment over chance, so the same fake name recurs across feeds. Two further checks establish that it is a *per-candidate* alias rather than a shared pool: 4,906 of 5,278 distinct aliases point at exactly one vehicle, and address rows carrying an alias are unlinked by name matching 72.6% of the time against a 19.5% baseline for the feed as a whole. The alias rows are precisely the rows name matching drops.

So the alias is learned on the title feed — where `vehicle_ref` reveals which candidate the odd row belongs to — then used to claim matching rows in the feeds that have no key of their own. Aliases pointing at more than one candidate are discarded. 8,067 aliases are recovered, covering two thirds of the candidates.

Pass 3 — confidence-weighted name matching for everything else. The matcher normalises the `SYN...` prefixes, blocks on a SymSpell-style delete neighbourhood (edit distance ≤ 2 , covering substitution, insertion, deletion and transposition) plus an explicit prefix index (covering truncation to an initial), and scores first and last name jointly. Date-of-birth agreement adds a bonus on the licence feed, never a penalty. Matches are softened into weights summing to 1 across competing candidates, so an ambiguous name splits its evidence rather than committing to a guess.

Rows that no pass claims are left unclaimed. An earlier version forced them on, using the 4-rows-per-candidate structure as a capacity constraint; coverage rose to ~99% but the address, licence and external signals all got *weaker*, because an unrelated name carries an unrelated state.

feed	linked	of	via alias	note
address	41,567	48,121	2,827	
licence	40,855	48,124	2,129	name + DOB
external	41,508	48,116	2,877	
work	24,131	24,131	0	this feed is not corrupted
title	47,581	48,108	—	via <code>vehicle_ref</code>
T1 updates	21,949	24,000	1,856	4,713 via <code>vehicle_ref</code>

3. Signal, and what carries none

Spearman rho of a recency-weighted DE share against the ordered T0 label:

feed	rho
current address only (blank end date)	+0.41
title	+0.37
address, all rows	+0.32
external	+0.29
licence	+0.15
work location	-0.07
four feeds pooled	+0.53

Two rows in that table decided features.

The work-location feed is complete, uncorrupted, and unrelated to the labels. It is excluded from the pooled score and kept only as its own near-zero feature block, so the model can confirm that rather than being forced to assume it.

The address record with a blank `effective_end_date` is where the person lives *now*, and that **single record out-predicts the entire address feed** (+0.41 against +0.32). It therefore gets its own feature block rather than being averaged into the history.

Recency dominates. Aggregations compared on the same resolved evidence (`model_selection.py`, section C):

aggregation	rho
exponential decay, 150-day half-life	+0.52
rank decay (0.5 per step back)	+0.51
newest record per feed, averaged	+0.48
hard window, last 180 days	+0.46
change-point, post-segment mean	+0.41
linear-in-time extrapolation	+0.39

The half-life sits on a broad plateau between 90 and 240 days; 150 days is used. That a change-point fit does worse says the evidence behaves like a recency-weighted mixture rather than a clean move date.

The record-type fields (`event_type`, `signal_type`, `source_type`) are almost entirely recency in disguise — on titles, `title_record` averages 1,729 days old, `ownership_change` 825, `record_update` 287 — so the date weighting already captures them. What they do add is that some categories carry two rows per candidate instead of one, a less noisy read of the same feed; those three categories are kept as features.

The observed tag state contributes independently and more weakly (rho +0.11): an out-of-state tag raises the score a little at any level of residency evidence. There is no interaction worth modelling.

4. Model

A **proportional-odds (ordinal) logistic model** over

```
review_not_warranted (0) < insufficient_evidence (1) < review_warranted (2)
 $P(y \leq k) = \text{sigmoid}(\theta_k - x.\text{beta}), \theta_0 < \theta_1$ 
```

fitted by L-BFGS with an L2 penalty ($C = 0.02$), implemented in `model.py`.

Why ordinal rather than a 3-way classifier: it encodes the banded transition structure found in section 1, and it estimates one direction vector plus two thresholds instead of three independent ones — which matters at 300 labelled candidates.

Every family was run on identical grouped folds, each with a hyperparameter sweep. The boosters were given two extra advantages so the comparison is not a straw man: the observed tag state was passed to CatBoost as a genuine nine-level categorical, and each booster was also run under the same ordinal decomposition the winner uses (`benchmark_models.py`, full table in `benchmark_output.txt`).

model	log loss	accuracy	macro F1	macro AUC
ordinal logistic	0.907	0.554	0.556	0.741
multinomial logistic	0.924	0.538	0.532	0.725
random forest	0.951	0.527	0.523	0.703
CatBoost (with categorical)	0.964	0.510	0.508	0.705
XGBoost	0.983	0.507	0.504	0.695
hist gradient boosting	0.998	0.503	0.501	0.691
LightGBM	1.047	0.496	0.495	0.682
class prior	1.096	0.355	0.175	0.500
LLM classifier (Claude Haiku 4.5)	1.151	0.497	—	0.659

Best configuration shown per family. Three results are worth stating plainly.

Every tree ensemble loses to plain logistic regression here, which inverts the usual expectation. Three reasons, and they compound: 600 training rows against 41 features is an order of magnitude less data than boosting wants; the signal is a smooth, near-monotone function of a single latent direction, which trees can only approximate as a staircase, spending variance to do it; and the ordinal constraint is information a multiclass tree cannot represent at all.

The gap is mostly calibration, not discrimination. CatBoost trails on accuracy by about four points but on log loss by 0.06 — tree ensembles are poorly calibrated out of the box, and calibration is one of the six scored criteria.

Handing the boosters the ordinal decomposition made them worse, not better (XGBoost $0.983 \rightarrow 1.037$, CatBoost $0.964 \rightarrow 1.145$). The decomposition trains two models where there was one, and at this sample size the extra variance costs more than the structure gains. The ordinal logistic gets the same structure for the price of two scalars.

Training rows. T0 and T1 are pooled into 600 rows with a phase indicator, so both phases train one model. Cross-validation groups by candidate so a candidate's T0 and T1 rows never straddle a fold.

41 features in five blocks:

- *evidence direction* — recency-weighted DE share of the four signal feeds at 90/150/365-day half-lives and unweighted; the state on the newest record; how many feeds end in DE; disagreement between feeds; per-feed direction and newest state; the three two-row record categories
- *current address* — direction, count, newest state and age of the open-ended address records
- *evidence sufficiency* — volume of linked evidence, records in the last 6 and 12 months, age of the newest record, share of records with no state, share of limited-quality records, share of expired or superseded records, identity-match confidence, number of distinct states seen
- *T1 updates* — volume, direction, direction of the newest update, direction of the title and address updates, and the change against prior evidence. Encoded as signed deviations that are exactly zero in the T0 phase, so the block contributes nothing before the updates exist rather than relying on imputation.
- *observed tag* — whether the tag on the candidate record is out of state

The four largest standardised coefficients are `t1_recupd_de` (+0.28), `t1_adr_de` (+0.26), `lic_credupd_de` (+0.22) and `adr_open_last_de` (+0.21). Three of the four come from blocks the first version of this model did not have.

5. Results

Grouped 5-fold cross-validation, 20 repeats, on all 600 labelled rows:

metric	model	baseline	first version
multi-class log loss	0.906	1.096 (class prior)	0.934
accuracy	0.558	0.355 (majority class)	0.527
macro F1	0.560	0.175	0.528
macro AUC	0.740	0.500	0.719
priority AUC for <code>review_warranted</code>	0.811	0.500	0.781

By phase — the model responds to the later evidence, as intended:

	log loss	accuracy	macro AUC	priority AUC
T0	0.931	0.520	0.718	0.794
T1	0.880	0.597	0.762	0.829

Confusion matrix (rows true, columns predicted, order not-warranted / insufficient / warranted):

```
[[136  61  16]
 [ 61  96  51]
 [ 15  61 103]]
```

The off-diagonal mass sits where the ordinal structure says it should: confusion is mostly with the adjacent class, and only 31 of 600 rows cross the whole scale.

Calibration. Predicted vs observed frequency in quintiles of predicted probability:

quintile	p(warranted) pred / obs	p(not warranted) pred / obs
1	0.05 / 0.01	0.07 / 0.11
2	0.13 / 0.17	0.17 / 0.14
3	0.24 / 0.22	0.30 / 0.33
4	0.41 / 0.42	0.47 / 0.42
5	0.65 / 0.67	0.73 / 0.78

No post-hoc recalibration is applied; the ordinal likelihood is already well calibrated on held-out folds, and with 300 labelled candidates a second fitted calibration layer would cost more in variance than it gains.

Class prior. A Kolmogorov-Smirnov test of the strongest feature, labelled subset vs all 12,000 candidates, gives $D = 0.047$, $p = 0.52$ — the development set behaves like a random sample, so its 35/35/30 class mix is used as the population prior rather than being reweighted.

6. Is the number honest?

Every figure above comes from cross-validation on the same 300 labelled candidates that also drove roughly forty design comparisons. That inflates them. Two checks quantify by how much.

Nested selection (`audit_honest.py`). The choice of feature blocks and regularisation is made *inside* a training split, and the chosen model is scored once on candidates it never saw. This is the unbiased estimate for the procedure, not just for its winner.

	log loss	accuracy	macro AUC	priority AUC
ordinary cross-validation	0.906	0.558	0.740	0.811
nested selection	0.913	0.549	0.737	0.809

The selection bias is 0.008 of log loss. The inner loop chose the same configuration in 27 of 30 folds, so the choice is stable rather than a coin flip. The unbiased new figure (0.913) still beats the first version's *optimistic* figure (0.934).

Paired bootstrap over candidates. Resampling the 300 candidates 4,000 times, the current feature set beats the first version by 0.022 of log loss, 95% interval $[+0.003, +0.040]$, better in 98.8% of resamples.

Treat 0.913 as the number to expect on the held-out evaluation cases.

7. Review priority

```
review_priority = p(review_warranted) + 0.5 x p(insufficient_evidence)
```

A case that warrants review takes full priority. A case the evidence cannot decide still needs a person to look at it — that is what `insufficient_evidence` means operationally — so it takes half. A case that clearly does not warrant review takes almost none. The value lies in $[0, 1]$ by construction, and it ranks marginally better than `p(review_warranted)` alone; the definition is chosen for what it means to a queue, not for the third decimal place.

8. Using the output

`case_predictions.csv` is the submission file, in the template's exact schema and row order.

`case_audit.csv` is a supporting file with the same 24,000 rows plus the evidence behind each score, so a reviewer can see why a case is where it is without reading the code:

- `latest_{address,licence,title,external}_{state,date}` — the newest record per feed
- `de_share_recent`, `de_share_all_time` — the direction of the evidence
- `evidence_records_linked`, `days_since_newest_record` — how much there is and how fresh
- `t1_update_direction` — toward DE / away from DE / mixed / no update
- `driver_summary` — signed contribution of each feature block, largest first
- `top_drivers` — the three individual features with the largest contribution

Signs read as: **positive pushes toward `review_warranted`, negative toward `review_not_warranted`**. Lead with `driver_summary`; individual coefficients are ridge-shrunk across collinear direction features, so a single feature's sign should not be read on its own, whereas the block sums are stable.

The output supports staff review. It is not a residency, fee, or enforcement determination, and `review_warranted` means "a person should look at this", nothing more.

9. What was tried and rejected

Each was judged on the same grouped cross-validation as the shipped model. Recording the failures is part of the answer: it is evidence the model sits on a plateau rather than at the first thing that worked.

idea	result
per-feed 90- and 365-day half-lives	log loss +0.001, no gain
splitting T1 updates by <code>record_action</code>	+0.000, noise
non-linear terms on the direction index	+0.002, worse
a "T1 title changed state" flag	-0.002, inside the noise
observed tag x direction interaction	+0.002, worse; the tag effect is additive
explicit old-versus-new trend per feed	+0.006, worse alone and in combination
self-training on the ~11,700 unlabelled cases	worse at every confidence threshold (0.55/0.65/0.75) and pseudo-label weight (0.02/0.05/0.15), from +0.002 to +0.165 log loss. Pseudo-labels sharpen the model's existing beliefs and wreck the calibration the scoring rewards.
forcing unclaimed rows on via the capacity constraint	coverage rose to ~99% but address, licence and external signal all got weaker
recovering the replaced address row from the gap it leaves in the timeline	not possible: consecutive address records chain end-to-start in 3 of 8,322 cases, and gaps run from -136 to +413 days. The records do not tile time.

10. Reproducibility

`predict.py` is deterministic. Run from a clean directory with the data at an arbitrary path (`00S_DATA_DIR`), it reproduces `case_predictions.csv` byte for byte, and `validate.py` passes all eleven format checks.

11. Limitations

- **300 labelled candidates.** Every metric carries roughly plus or minus 0.03 of sampling noise, and the selection bias on top of that is quantified in section 6. The half-life and feature blocks were chosen on the flat part of their curves rather than at the cross-validation optimum. That said, more labels would not buy much: the learning curve (`audit_learning_curve.py`, held-out set held at 75 candidates throughout) has already flattened.

training candidates	40	70	105	145	190	225
log loss	0.976	0.943	0.928	0.922	0.916	0.914

The marginal value of 50 extra labelled candidates falls from +0.054 log loss early on to +0.003 at the top of the range. The binding constraint is the noise in the evidence itself, not the size of the label set.

- **About 13% of evidence rows remain unattributable** after all three resolution passes. The alias route recovers the replaced row only where the candidate's own vehicle gave a clean 3 + 1 split, which is two thirds of candidates. For the rest, nothing in the package identifies the row.
- **The signal itself is noisy.** Even with perfect resolution a candidate has roughly fourteen state-bearing records split between two states, so the latent direction can only be estimated to limited precision.
- **About 350 vehicles carry two candidates' histories** (343 have eight title rows, six have nine). They are split by name cluster, but where both clusters are weak the split is uncertain.
- **The noise model is inferred, not documented.** It is verified three ways — vehicle grouping, date-of-birth grouping, and the cross-feed alias enrichment — but a fourth corruption mode that none of those views exposes would not have been caught.